

Progress Test 1.B — Elec-S1

1.B · From charge to electric current
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion (1 mark)

The capacity of a phone battery is $Q = 4000\text{ mA h}$, i.e. the charge it delivers when a current of 4000 mA flows for one hour. Express Q in coulombs. Give the exact result.

Marking

Exercise 2. Microscopic expression of the current (5 marks)

A conductor of cross-sectional area S contains n mobile charge carriers per unit volume, each of charge q , all moving with the drift velocity v along the axis of the conductor. Express the charge dq that crosses S during dt (explain your reasoning), deduce the current i , and check that your result is dimensionally consistent, stating the SI unit of n .

Marking

Exercise 3. Current and ammeters

(3 marks)

In a copper wire carrying a current, the conduction electrons move:

- ☐ in the conventional direction of the current;
- ☐ in the direction opposite to the conventional direction of the current;
- ☐ at a speed close to that of the electric signal;
- ☐ only when the wire is heated.

An ammeter is connected:

- ☐ in parallel across the component, and has a very large internal resistance;
- ☐ in series in the branch, and has a very small internal resistance;
- ☐ in series in the branch, and has a very large internal resistance;
- ☐ in parallel across the component, and has a very small internal resistance.

Which of these units is one of the seven SI base units?

- ☐ coulomb (C) ☐ volt (V) ☐ ampere (A) ☐ ohm (Ω)

Exercise 4. Counting electrons

(5 marks)

A steady current i flows through a wire during a time Δt . Determine the number N of electrons that cross a cross-section of the wire during Δt .

$i = 3.2 \text{ mA}$	$\Delta t = 5.0 \text{ s}$	$e = 1.6 \times 10^{-19} \text{ C}$
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Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1
⑤	
0	1

— End of the test —